

mm

cm

UPN 50

Geometria

$h = 50 \text{ mm}$

$b = 38 \text{ mm}$

$t_f = 7 \text{ mm}$

$t_w = 5 \text{ mm}$

$r_1 = 7 \text{ mm}$

$r_2 = 3,5 \text{ mm}$

$y_s = 13,7 \text{ mm}$

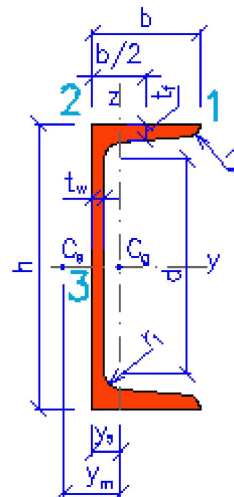
$y_m = 25 \text{ mm}$

$d = 20,8 \text{ mm}$

$G = 5,59 \text{ kg.m}^{-1}$

$A_L = 0,23 \text{ m}^2$

$A = 712 \text{ mm}^2$



Propriedades da seção

Eixo y

Eixo z

$I_y = 2,65 \text{E}+5 \text{ mm}^4$

$I_z = 9,09 \text{E}+4 \text{ mm}^4$

$W_y = 1,06 \text{E}+4 \text{ mm}^3$

$W_{z1} = 3740 \text{ mm}^3$

$W_{z2} = 6640 \text{ mm}^3$

$W_{y,pl} = 1,30 \text{E}+4 \text{ mm}^3$

$W_{z,pl} = 6890 \text{ mm}^3$

$i_y = 19,3 \text{ mm}$

$i_z = 11,3 \text{ mm}$

$S_y = 6490 \text{ mm}^3$

Deformação e enrugamento

$I_w = 2,84E+7 \text{ mm}^6$	$I_t = 1,11E+4 \text{ mm}^4$
$i_w = 5,95 \text{ mm}$	$i_{pc} = 22,4 \text{ mm}$